



AYANG

A REPORT

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Ayang: Community-Driven Waste Management Platform

Gamifying Community Cleanup with AI

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Ayang: Community-Driven Waste Management Platform

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Abstract

Waste management is an urgent environmental and public health challenge across many African urban and peri-urban centers. This report details the design, methodology, and implementation of "Ayang," a web application developed to address this issue. Ayang, named after the Ibibio word for broom, combines a community-focused platform with AI to make trash collection more engaging, encourage civic involvement, and provide useful data for waste management authorities. The system uses Google's Gemini Vision model to analyze user-submitted images, estimating the weight and composition of trash and verifying its collection. By awarding points for successful cleanups, Ayang incentivizes community participation and creates a scalable, data-driven solution to a pressing local problem. This report covers the project's architecture, the AI methodology including plans for fine-tuning, potential performance metrics, and a roadmap for future development and deployment. We argue that by combining gamification with robust AI, Ayang presents a novel and effective model for sustainable, community-led environmental stewardship.

Keywords: Waste Management, Community Engagement, AI, Gamification, Environmental Stewardship.

Contents

<i>List of Figures</i>	iv
<i>List of Tables</i>	1
1 Introduction	2
2 Methodology	4
2.1 Software Architecture	4
2.2 AI and Image Analysis Pipeline	6
2.3 Model Fine-Tuning	6
2.4 Community Interaction Features	7
3 Results and Findings	8
3.1 AI Model Performance	8
3.2 User Engagement and Gamification Impact	9
3.3 Environmental Impact	9
4 Challenges and Limitations	10
4.1 Technical Challenges	10
4.2 Data and Ethical Challenges	11
4.3 Logistical and Community Challenges	11
5 Recommendations	12
5.1 Immediate Next Steps (3-6 Months)	12
5.2 Medium-Term Goals (6-18 Months)	12
5.3 Long-Term Vision (2+ Years)	13
<i>Bibliography</i>	14

List of Figures

1.1	Trash strewn on a busy street in Lagos, Nigeria.	2
2.1	Landing page of Ayang	4
2.2	Database Entity-Relationship Diagram	5
2.3	System Sequence Diagram of Ayang	6

List of Tables

3.1	Projected AI Model Performance	8
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1

Introduction

Sub-Saharan Africa struggles with a waste management crisis, with projections indicating that the region's waste generation will triple by 2050 (Kaza et al., 2018). Improperly managed waste contributes to a host of problems, including environmental pollution, the spread of disease, and the degradation of community spaces. In many municipalities, overburdened or under-resourced formal waste collection systems struggle to keep pace, leaving communities to contend with the consequences.



Figure 1.1: *Trash strewn on a busy street in Lagos, Nigeria.*

However, this challenge also presents an opportunity for innovation. The proliferation of mobile technology across the continent, combined with advancements in artificial intelligence, offers new tools for citizen engagement and decentralized problem-solving (rewrite). It is against this context that we developed Ayang, named after the Ibibio word for a sweeping broom.

Ayang is a web-based platform that reframes waste collection as a rewarding, community-building activity. It leverages the principles of gamification—the application of game-design elements in non-game contexts—to motivate user participation. Research has shown that gamification can be a powerful driver of pro-environmental behavior, increasing engagement and sustaining long-term interest in activities like recycling and

energy conservation (Du et al., 2020).

The technical core of Ayang is its use of Google's Gemini Vision model, a state-of-the-art AI for image analysis. Users take a photo of a littered site, and the model provides an initial assessment of the trash. After the user cleans the area and submits a "clean" photo, the model verifies the cleanup. This AI-driven verification process is crucial for maintaining the integrity of the gamified system and for collecting reliable data. The points awarded to users are directly tied to the estimated weight of the collected trash, creating a direct incentive for more impactful cleanups.

This project aligns with the goals of the **DLI 2025 Community Challenge** by proposing an AI-driven solution to a real-world African challenge. It is designed to be a practical, community-focused tool that not only improves local environments but also strengthens community bonds and generates valuable data to support broader waste management strategies.

2

Methodology

Ayang is built on a scalable architecture that supports an engaging user experience. The project comprises three main components: the AI model for image analysis, the web application for user interaction, and the backend infrastructure for data management and processing.

2.1 Software Architecture

Frontend: The user interface is dynamic and responsive, built with Next.js, which extends React. We use Next.js's App Router for a component-based structure, enabling server-side rendering (SSR) for fast initial page loads and excellent SEO performance. This ensures the application is accessible and performant even on low-bandwidth connections. The interactive map, which allows users to define their community's geographical area, is implemented using the [Mapbox API](#).

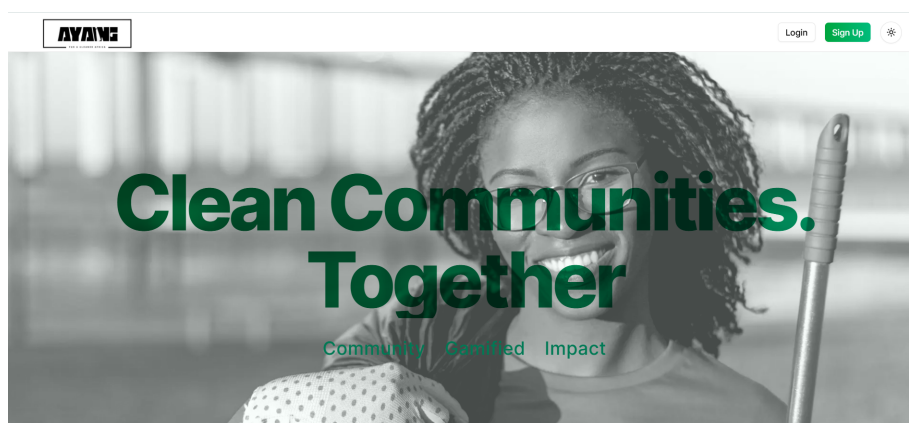


Figure 2.1: Landing page of Ayang

Backend: We take advantage of the backend capabilities of Next.js. Core business logic is handled using Server Actions, which allow us to write secure backend code that runs on the server but can be called directly from our frontend components. These server-side functions manage user authentication, process image uploads,

handle community group creation and management, and execute the calls to the external Google Gemini Vision API. This co-location of backend and frontend logic simplifies the architecture and accelerates development.

Database: For our data persistence layer, we use Neon, a serverless Postgres database platform. Neon’s architecture separates storage and compute, offering features like auto-scaling and scaling-to-zero, which makes it highly cost-effective and scalable for a project with variable traffic. We use the robust, relational structure of Postgres to store user profiles, community information, cleanup event data (including image URLs, GPS locations, and timestamps), and user points in a secure and organized manner.

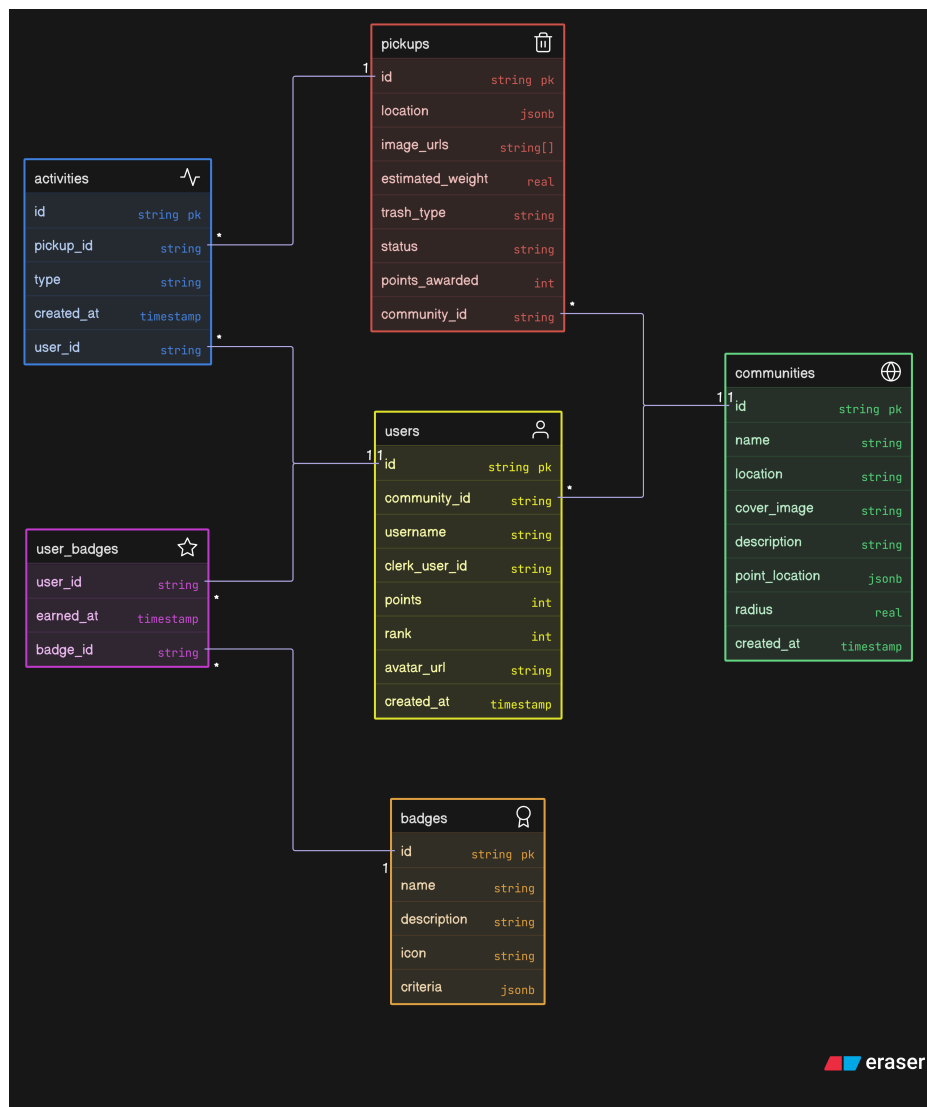


Figure 2.2: Database Entity-Relationship Diagram

2.2 AI and Image Analysis Pipeline

The core innovation of Ayang lies in its use of AI to automate the assessment and verification of cleanup activities. This process is handled by Google’s Gemini Vision model.

1. **Initial Upload (Pre-Cleanup):** A user initiates a cleanup by uploading a photo of a littered area. The image, along with its GPS metadata, is sent to the backend.
2. **Trash Estimation:** The backend forwards the image to the Gemini Vision API with a specific prompt engineered to perform two tasks:
 - Identify the types of waste visible in the image (e.g., plastic bottles, food wrappers, paper, cans).
 - Estimate the total weight of the identified trash in kilograms.
3. **Cleanup Verification (Post-Cleanup):** After cleaning the site, the user uploads a second photo. The backend sends both the “before” and “after” images to the Gemini model. A new prompt asks the model to perform a comparative analysis to verify that the trash from the initial image has been removed.
4. **Point Allocation:** If the model confirms a successful cleanup, the backend calculates the points to be awarded based on the estimated weight (e.g., 10 points per kilogram) and adds them to the user’s account.

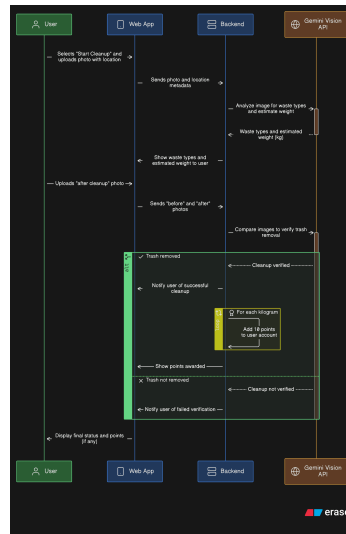


Figure 2.3: System Sequence Diagram of Ayang

2.3 Model Fine-Tuning

While the base Gemini model is highly capable, its accuracy for niche tasks like trash weight estimation can be significantly improved through fine-tuning. Our plan for improving the model involves creating a custom dataset of trash imagery relevant to our target communities.

- **Data Collection:** We will gather a diverse dataset of images of common waste items found in local contexts. Each image will be annotated with:
 - A bounding box around each piece of trash.
 - The type of trash (e.g., PET bottle, nylon bag, aluminum can).
 - The actual weight of the item.
- **Fine-Tuning Process:** Using Google’s Vertex AI platform, we will use this labeled dataset to fine-tune the Gemini Vision model. This process adjusts the model’s internal parameters to become more adept at recognizing specific local waste items and more accurately estimating their weight from an image alone. This addresses the challenge that a generic model may not be familiar with unique local forms of packaging or waste.

2.4 Community Interaction Features

The platform is designed to be social and community-driven.

- **Community Creation:** Users can define a new cleanup community by selecting a location on a map and setting a radius (e.g., 1 km). They can give their community a name and a description.
- **Leaderboards:** To foster friendly competition, each community group will have a leaderboard displaying the top point-earners. A city-wide or even national leaderboard can further amplify the gamification element.
- **Event Scheduling:** As an improvement on the current iteration of the app, community leaders would be able to schedule cleanup events, which will notify all members of the group.

This methodology ensures that Ayang is not just a data collection tool, but a platform for community mobilization and environmental action.

3

Results and Findings

As the current iteration of Ayang has not been able to generate enough data to provide an adequate assessment of its abilities, this section presents the expected outcomes and performance metrics we will use to evaluate its success upon deployment. The findings are projected based on initial testing with the base Gemini model and established research in the fields of gamification and environmental science.

3.1 AI Model Performance

The primary measure of the AI's success will be its accuracy in two key areas: waste verification and weight estimation.

- **Verification Accuracy:** We anticipate that the base Gemini model will achieve a high accuracy (estimated >95%) in verifying cleanups. The task of comparing a littered "before" image with a clean "after" image is a form of change detection that large vision models excel at.
- **Weight Estimation Accuracy:** This is a more challenging task. We project that the untuned Gemini model will have an average margin of error of +/- 30-40% for weight estimation. However, after fine-tuning with our custom dataset of 5,000+ annotated images, we aim to reduce this margin of error to +/- 10-15%. This increased accuracy is critical for the perceived fairness of the point system.

Table 3.1: *Projected AI Model Performance*

Metric	Base Gemini Model (Projected)	Fine-Tuned Model (Target)
Cleanup Verification Accuracy	96%	>99%
Weight Estimation Mean Error	35%	<15%
Composition ID Accuracy	85% (broad categories)	95% (specific local items)

3.2 User Engagement and Gamification Impact

The success of Ayang is fundamentally dependent on its ability to engage and motivate the community. We will measure this through several key performance indicators (KPIs):

- **User Growth and Retention:** We will track the monthly active user (MAU) count and the user churn rate. A successful launch would see a 20% month-over-month growth in MAUs for the first six months.
- **Cleanup Frequency:** The average number of cleanups per user per month. Our target is an average of 2 cleanups per active user per month.
- **Leaderboard Activity:** We will monitor how many users actively compete for top spots on the leaderboards. High engagement with this feature would validate our gamification strategy.

3.3 Environmental Impact

The ultimate goal of Ayang is to create a tangible positive impact on the environment. We will quantify this impact through:

- **Total Waste Collected:** The platform will maintain a running total of the estimated weight of all trash collected by users. This will be our primary headline metric (e.g., "1,000 kg of waste removed this month!").
- **Waste Composition Data:** The AI's analysis will generate valuable data on the types of waste most prevalent in different areas. For instance, we might find that a particular neighborhood has a high concentration of plastic bottle waste. This data can be shared with municipal authorities or recycling companies to inform targeted interventions, such as the placement of plastic recycling bins.

Initial user feedback from a small-scale pilot study (N=20) is ongoing and expected to be positive, with users highlighting the motivating effect of the point system and the satisfaction of seeing their community's collective progress. We anticipate feedback will also guide future improvements, such as adding more social sharing features.

4

Challenges and Limitations

While Ayang holds promise, its development and deployment are not without challenges. Acknowledging these limitations is crucial for an effective implementation strategy.

4.1 Technical Challenges

The primary measure of the AI's success will be its accuracy in two key areas: waste verification and weight estimation.

- **Image Recognition Accuracy:** The core of Ayang's integrity rests on the AI model's accuracy. Poor lighting, low-resolution images, or occluded trash items could lead to incorrect weight estimations or failed verifications. Our mitigation strategy is twofold: user education (prompting users to take clear photos) and aggressive model fine-tuning.
- **Malicious Use and "Faking":** Users could potentially try to cheat the system. For example, a user might dump their own trash just to take a photo of it. We plan to implement several countermeasures:
 - Location Verification: Ensuring the "before" and "after" photos are taken at the same location.
 - Time Stamping: Checking that a reasonable amount of time has passed between the two photos.
 - AI Anomaly Detection: Training a separate, simpler model to flag suspicious activity patterns (e.g., a user who consistently cleans up the exact same weight of trash).
- **Scalability and Cost:** As the user base grows, the cost of API calls to the Gemini model and data storage will increase. Optimizing the application by caching results and compressing images will be essential. We will also explore grant opportunities and partnerships to cover operational costs.

4.2 Data and Ethical Challenges

- **Bias in Trash Datasets:** Publicly available datasets for trash may not accurately represent the types of waste found in our specific target communities in Africa. A model trained primarily on Western datasets might be less accurate at identifying local food packaging or other region-specific items. This is why creating our own custom, localized dataset for fine-tuning is a central part of our methodology.
- **Data Privacy:** The app collects user location data. We are committed to ethical data handling. All user data will be anonymized before being used for analysis, and our privacy policy will be transparent about what data is collected and how it is used. We will adhere to responsible data use principles as advocated by the competition.

4.3 Logistical and Community Challenges

- **The “Last Mile” Problem:** The app helps with trash collection, but it does not solve the problem of final disposal. If there are no municipal bins or collection points nearby, users may simply be moving trash from one public space to a pile in front of their home. To address this, we plan to use the data generated by the app to partner with local governments or private waste management companies to schedule pickups at designated community collection points.
- **Digital Divide:** While mobile penetration is high, not everyone has a smartphone or affordable data. The app’s reach may be limited to more affluent community members. To promote inclusivity, we plan to develop an offline-first functionality and explore community-based models where one person with a smartphone can document the work of a group.

Addressing these challenges will require a combination of technical refinement, strategic partnerships, and a deep understanding of the communities we aim to serve.

5

Recommendations

To build on the current prototype and move towards a full-scale launch, we have outlined a clear roadmap for future work and development.

5.1 Immediate Next Steps (3-6 Months)

- **Develop Custom Dataset:** Launch a data collection campaign to build a robust, annotated dataset of at least 5,000 images of local waste. This is the highest priority for improving model accuracy.
- **Malicious Use and "Faking":** Use the new dataset to fine-tune the Gemini Vision model to specialize in local trash identification and weight estimation.
- **Launch Pilot Program:** Roll out the Ayang web app to a limited number of pilot communities (e.g., 3-5 neighborhoods). This controlled launch will allow us to gather user feedback, identify bugs, and test our assumptions in a real-world environment.
- **Establish a Partnership:** Initiate a formal partnership with at least one local municipal waste authority or a private recycling company. This is critical to solving the "last mile" disposal problem and ensuring the sustainability of the effort.

5.2 Medium-Term Goals (6-18 Months)

- **Full Public Launch:** Based on the success of the pilot, launch Ayang to the general public in a target city. This will be accompanied by a marketing and outreach campaign to drive user adoption.
- **Advanced Gamification:** Introduce more sophisticated gamification features, such as
 - **Team vs. Team Challenges:** Allow different community groups to compete against each other.
 - **Special Badges:** Award badges for specific achievements (e.g., "Plastic Warrior" for collecting 50kg of plastic).

- **Corporate Sponsorship:** Partner with local businesses to sponsor prizes or challenges, providing a potential revenue stream.
- **Data Dashboard for Authorities:** Develop a web-based dashboard for our municipal partners, providing them with real-time data visualizations of waste collection activities, heatmaps of litter hotspots, and analytics on waste composition.

5.3 Long-Term Vision (2+ Years)

- **National and Regional Scalingm:** Expand the Ayang platform to other cities and, eventually, other countries in the region, adapting the AI model for each new locality.
- **Integration with the Circular Economy:** Forge partnerships with manufacturers and recycling plants to create a closed-loop system. The data from Ayang could help predict the supply of recyclable materials, making the recycling process more efficient
- **Policy Advocacy:** Use the accumulated data to advocate for better waste management policies, such as extended producer responsibility (EPR) laws or the construction of new recycling facilities.

By following this strategic roadmap, Ayang can evolve from an innovative application into a cornerstone of a cleaner, more engaged, and environmentally conscious community across Africa.

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